Excitation Regime, Not the Degree of Freedom: Identifiability in Grey-Box Digital Twinning of an Omnidirectional Soccer Robot

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***Abstract*—Robot soccer teams in the RoboCup Small Size League develop and validate control software largely in simulation, where simplified physics leaves a reality gap to the field. We build a behavioral digital twin of an omnidirectional soccer robot as three first-order-plus-dead-time channels, one per degree of freedom, identified mainly from closed-loop competition logs that teams already collect. Fidelity is evaluated in free run, over multiple horizons, with a re-anchored position error, against an ideal-kinematics baseline and rSim, an ODE-based league simulator. The translational channels reach roughly 87% hold-out fit, and the twin outperforms the simulator on translation-dominated windows with a large, statistically gated effect. The central finding concerns yaw: the same model structure fails every acceptance gate when fitted to closed-loop logs, yet passes all pre-registered gates when fitted to a small dedicated open-loop collection. The recovered gain is near unity, which we read as evidence that the inflated closed-loop estimate reflects correlation bias rather than plant gain, and 94% of commanded yaw-rate samples in the logs fall below the robot's actuation threshold. We argue that identifiability claims drawn from operational logs are claims about the available excitation, not about the robot.**

***Index Terms*—digital twin, system identification, grey-box modeling, omnidirectional robot, robot soccer, sim-to-real, fidelity evaluation**

I. Introduction

In the RoboCup Small Size League (SSL), teams of fast omnidirectional four-wheeled robots are coordinated by software that is developed and validated largely in simulation. The reference simulation tools of the league implement a simplified omnidirectional wheel model on top of a rigid-body ODE engine [1], reinforcement-learning frameworks for robot soccer inherit that engine [2], and recent motion-planning policies for the SSL are trained inside these environments [3]. Because the simulated dynamic response deviates from the real robot, behaviors validated in simulation degrade on the field — the reality gap.

A higher-fidelity behavioral twin narrows this gap, but building one usually presumes either a detailed physical model or a dedicated excitation campaign with robot and field access. Competition logs are the data teams already have: body-frame velocity commands and vision poses recorded at $\sim$60 Hz during closed-loop play. A twin identified from such logs would be cheap and reproducible — at the price of an identifiability question, since the data are collected under feedback and were never designed to excite the plant.

Digital twins of mobile robots validate fidelity behaviorally but obtain their models from white-box physics or purpose-collected data: simulation models of omnidirectional platforms are tuned on dedicated real trajectories [4], [5], identification pipelines design excitation sequences per parameter group [6], grey-box and real-time estimation for twins assume informative operating data [7], [8], and offline sim-to-real calibration presumes the pre-collected data excite the parameters of interest [9]. Whether a behavioral twin can be identified from closed-loop *operational* logs — and which degrees of freedom such logs support — has, to our knowledge, not been characterized.

This paper models an omnidirectional SSL robot as three decoupled first-order-plus-dead-time (FOPDT) channels, one per degree of freedom (DOF), and evaluates fidelity in free run over multiple horizons with a re-anchored position error, against an ideal-kinematics baseline and against rSim, the ODE engine of the rSoccer environment [2]. The translational channels identify cleanly from competition logs (hold-out fit $\sim$87%) and the twin outperforms rSim on translation-dominated windows with a large, statistically gated effect. The yaw channel does not: the *same* model structure fails every acceptance gate on closed-loop data and passes all pre-registered gates on a small dedicated open-loop collection, with the discrepancy explained by the robot's own actuation threshold — 94.3% of commanded yaw-rate samples in the logs fall below it — rather than by the model. Identifiability, in this data regime, is a property of the excitation, not of the DOF. The result is thus a contribution to robot modeling and, at the same time, a higher-fidelity behavioral plant for the robot simulation and visualization tools on which robot soccer development relies.

Contributions. (C1) A per-DOF grey-box (FOPDT) identification methodology for an omnidirectional robot from *closed-loop* competition logs, with per-DOF hold-out validation. **(C2)** A free-run, multi-horizon, re-anchored fidelity protocol with motion-stratified windows and a statistically gated verdict (paired non-overlapping Wilcoxon, moving-block bootstrap, effect size). **(C3)** A validated comparison showing the twin attains higher *translational* fidelity than rSim on translation-dominated windows. **(C4)** A characterization of yaw identifiability as a function of the *excitation regime*: the same FOPDT structure fails on closed-loop competition logs and passes every pre-registered gate on a dedicated open-loop collection, the contrast pointing to the inflated closed-loop gain as correlation bias rather than plant gain.

II. Related Work

a) Behavioral digital twins of mobile robots: Digital twins of mobile platforms are typically validated behaviorally, by comparing simulated and real trajectories rather than parameters [10], [11]. On the construction side, grey-box identification has been evaluated as the model-generating module of a twin, and found sensitive to the informativeness of its input data [7].

b) Identification of omnidirectional robots: Work on omnidirectional platforms assumes dedicated experiments: the Otbot project designs a three-step identification sequence, one experiment per parameter group [6]; simulation models of highly dynamic omnidirectional logistics robots are tuned manually against motion-capture trajectories, requiring distinct parameter sets per trajectory [4], or tuned automatically by black-box optimization, where a few high-impact parameters outperform tuning many insensitive ones [5]. The FOPDT structure adopted here per DOF is the classic first-order-plus-dead-time process model of control practice [12].

c) Simulator calibration from offline data: Closest in spirit to using logs as-is, DROPO estimates domain-randomization distributions from an offline, pre-collected dataset [9]. Such approaches presume the offline data excite the parameters being estimated — the assumption this paper interrogates.

d) SSL simulators and league context: The SSL's reference simulator, grSim, adopts a simplified dynamic model of the omni wheels for real-time performance [1]; the rSoccer framework derives its rSim engine from grSim for reinforcement-learning workloads [2], and planning policies are trained in such environments [3]. A Simscape multibody (white-box) twin of an SSL robot reports trajectory RMSE of 150–240 mm on game scenarios [13]. League practice routinely records vision poses, odometry, and onboard velocities [14], and SSL firmware converts body commands into per-wheel velocity loops [15] — the actuation chain behind the per-wheel dead-zone analyzed in Sec. IV-D.

e) Closed-loop identification: That feedback correlates the input with unmeasured disturbances, breaking methods that work in open loop, is classic [16]–[18], as is the role of persistent excitation in prediction-error estimation [19].

f) Gap: No prior work, to our knowledge, identifies a behavioral twin of an omnidirectional robot from closed-loop *operational* logs and characterizes, on the same platform and model structure, which degrees of freedom that regime can and cannot identify.

III. Method

A. Per-DOF FOPDT twin

We model the robot as three decoupled single-input single-output channels, one per degree of freedom (DOF) $i \in \{v_x, v_y, \omega\}$, each a first-order-plus-dead-time (FOPDT) transfer function from commanded to attained body velocity,

$$G_i(s) \;=\; \frac{K_i\, e^{-T_{di}s}}{\tau_i s + 1}, \tag{1}$$

with per-DOF gain K_i, time constant τ_i, and transport delay T_{di}. The body-frame commands v_x, v_y (longitudinal/lateral, $\mathrm{m/s}$) and ω (yaw rate, $\mathrm{rad/s}$) are the inputs; the outputs are the attained body velocities. The full parameter vector has fixed order $[K_{vx}, \tau_{vx}, T_{dvx}, K_{vy}, \tau_{vy}, T_{dvy}, K_\omega, \tau_\omega, T_{d\omega}]$ (nine parameters). This is a *behavioral* (observational) model: it targets the input-to-pose response, not a white-box description of wheel/contact mechanics.

B. Discretization and pose integration

The twin is simulated on a uniform grid with step $\Delta t = 1/60\,\mathrm{s}$: each DOF's command is delayed by a zero-order-hold (ZOH) integer shift of $m_i = \lceil T_{di}/\Delta t \rceil$ samples (zero before the segment start), passed through the exact ZOH first-order update $v_i[k] = v_i[k-1] + (1 - e^{-\Delta t/\tau_i})(u_i^d[k] - v_i[k-1])$, and scaled to the attained body velocity $\hat{v}_i[k] = K_i\, v_i[k]$ (instantaneous for $\tau_i \to 0$). The body velocity is rotated through the current heading and integrated to the field frame by the trapezoidal rule; an independent reconciliation of this plant against the high-fidelity Simulink model agrees to $\leq 6\,\mathrm{mm}$ in position and $\leq 9 \times 10^{-4}\,\mathrm{rad}$ in heading on a reference event.

C. Data and pre-processing

The data are closed-loop competition logs of a single omnidirectional 4-wheeled SSL robot, decoded from the league protocol stream into per-event body-frame commands and field-frame pose at $\sim$60 Hz (heading unwrapped). Measured body velocity, used only for identification and for the initial condition, is a Savitzky–Golay derivative of the vision position rotated into the body frame. The native grid is irregular (command bursts); identification resamples to $\Delta t = 1/60\,\mathrm{s}$, while the fidelity metric uses time-based windows on the native grid. The dataset comprises 46 events across four motion labels (`front_to_back`, `side_to_side`, `shoot_to_goal`, `square`); a frozen list of 10 events is held out and never used for fitting.

D. Identification

Each translational DOF is identified independently as a P1D (FOPDT) model by prediction-error minimization, with an event-level hold-out (seed 42). The excitation diagnostic selects the cleanest source label per DOF: v_x from `front_to_back` and v_y from `side_to_side` (command-to-velocity cross-correlation ≈ 0.99). The yaw channel is treated separately: a direct P1D fit on the highest-rotation closed-loop label fails validation (Sec. IV-D), so we identify ω from a dedicated open-loop collection instead, reporting both attempts for contrast.

E. Dedicated open-loop yaw excitation

The dedicated dataset is single-session and single-robot with $v_x = v_y = 0$ throughout: an alternating-sign staircase of commanded yaw rates $\{0.25, 0.5, 1, 2.5, 5\}\,\mathrm{rad/s}$, $\sim 10\,\mathrm{s}$ per step (ten events). The two lowest amplitudes never engage the motors (zero commanded PWM) and serve only to characterize the per-wheel dead-zone (Sec. IV-D); the remaining

six events form the linear regime, split event-wise into four training and two hold-out events under the same pipeline as v_x, v_y. Given the small sample, the acceptance criteria — physical bounds, hold-out fit $\geq 60\,\%$, and the multi-horizon $\mathrm{SE}(2)$ gate against the closed-loop placeholder — were fixed and committed *before* estimation, guarding against post-hoc adjustment of criteria to the result.

F. Free-run replay and multi-horizon metric

Fidelity is measured in *free-run*: from the true pose at a window anchor t_0 the twin is driven only by the command stream, never re-anchored inside the window (no teacher forcing, which would measure single-step noise and inflate apparent fidelity). Segments are cut into overlapping windows at a fixed stride, each re-anchored to the true $\mathrm{SE}(2)$ pose at t_0 and rolled out to $H_{\max} = 90$ samples (1.5 s). We report, *separately*, the Euclidean position error $e_{\mathrm{pos}}(w, h)$ and the wrapped absolute heading error $e_{\mathrm{ang}}(w, h)$ at $h \in \{6, 12, 24, 42, 60, 90\}$ samples ($\{0.1, 0.2, 0.4, 0.7, 1.0, 1.5\}$ s), with reference horizon $h_{\mathrm{ref}} = 24$ (0.4 s). e_{pos} is the claim of this paper; e_{ang} is reported but bounded by the population of rotation-rich windows available (Sec. IV-B), independently of whether ω is identified.

a) Motion stratification: Each window is binned by its rotational fraction $r = a_{\mathrm{rot}}/(a_{\mathrm{rot}} + a_{\mathrm{trans}})$, where a_{trans} and a_{rot} accumulate translational command magnitude and yaw rate scaled by the robot radius. Windows are *translation-dominated* ($r < 0.2$), *mixed* ($0.2 \leq r \leq 0.5$), or *rotation-dominated* ($r > 0.5$).

b) Statistical gate: Curves use all (dense) windows; inference uses only *non-overlapping* windows (stride $\geq H_{\max}$) for a paired one-sided Wilcoxon signed-rank test, plus a moving-block bootstrap (block $\sim H_{\max}$) for the gap confidence interval, with any sign/order divergence flagged; effect size is the rank-biserial correlation of the paired differences. A bin receives a GO verdict only if, at all $h \geq h_{\mathrm{ref}}$, the gap is positive and non-decreasing, $p < \alpha$ ($\alpha = 0.05$), the effect size ≥ 0.2, and the bootstrap CI excludes zero, subject to $N_{\min} = 10$ independent windows; below $N_{\min}$ a bin is INDETERMINATE, which is the case for the mixed and rotation-dominated bins throughout (Sec. IV-B).

c) Horizon grid and anchor convention: All results use the $\mathrm{SE}(2)$ (full-pose) re-anchor above, but two independently implemented harnesses with different horizon grids: the fidelity results (Sec. IV-C) use the grid given above, while the per-DOF promotion gates (Secs. IV-A and IV-D) reuse an earlier campaign harness with $h \in \{0.5, 1, 2, 2.5, 4\}$ s. Each result is reported on its native grid, flagged at each point, rather than recomputed.

G. Baselines

We compare the twin against two baselines under the *same* windows, horizons, and gate. The first is an *ideal-kinematics* plant in which the attained body velocity equals the command instantaneously (no gain, lag, or delay), integrated by the same trapezoidal scheme. The second is **rSim**, the ODE-based physics engine of the rSoccer environment, driven by the same body-frame commands with motion control disabled so that the engine dynamics are isolated. rSim provides no velocity setter at reset (the robot starts from rest), so two initial-condition fairness modes are used: **Mode A** (primary) warms the engine up over $L_w = 15$ samples with the real pre-anchor commands and then applies a rigid $\mathrm{SE}(2)$ re-anchor at t_0, mirroring how the twin is seeded with the real body velocity; **Mode B** (robustness) uses no warm-up and discards the first 15 samples for *both* plants. The twin's transport delay has no counterpart in rSim, which is precisely where the twin is expected to win at short horizons.

TABLE I
IDENTIFIED PER-DOF FOPDT PARAMETERS AND HOLD-OUT FIT. ALL THREE CHANNELS ARE IDENTIFIED (NO PLACEHOLDER); ω USES A DEDICATED OPEN-LOOP ROTATION-ONLY COLLECTION (SEE SEC. IV-D), NOT THE CLOSED-LOOP COMPETITION LOGS USED FOR v_x, v_y.

DOF	K	τ (s)	T_d (s)	hold-out fit
v_x	1.119	0.072	0.069	86.7 %
v_y	1.052	0.127	0.056	87.1 %
ω	1.002	0.083	0.089	71.1 %

IV. RESULTS

A. Identified model

Table I reports the per-DOF parameters. Both translational channels identify as fast FOPDT models with hold-out fit 86.7% (v_x) and 87.1% (v_y); under the multi-horizon $\mathrm{SE}(2)$ gate against the previous (slower) parameters they improve every event-by-horizon cell (v_x: $20/20$; v_y: $65/65$), roughly halving the median position error at 2.5 s (v_y: 0.141 vs. 0.334 m). The yaw channel ω is also identified, from a separate open-loop dataset, with hold-out fit 71.1%; the reasons for the separate data source, and the contrast with the failed closed-loop attempt, are in Sec. IV-D. All three channels are deterministic bit-for-bit under re-execution (identical seed, data, and toolbox version): re-running the identification pipeline reproduces all nine parameters to 0.0 absolute difference.

B. Motion distribution of the hold-out

The hold-out windows are overwhelmingly translation-dominated: of 265 dense windows (49 independent), 264 fall in the translation bin (99.6%), one is rotation-dominated, and none are mixed. This conditions the analysis: translational fidelity is the statistically powered claim, while the rotation and mixed bins lack the population to support a verdict. This population is unaffected by the yaw model: re-running the bin classification and the twin-vs-ideal gate after ω was identified (Table I) reproduces the same distribution ($264/1/0$) and leaves mixed and rotation-dominated NO POWER (N_{indep}=0), now for lack of rotation-rich competition windows rather than for lack of a yaw model. The rotational fidelity of the twin therefore remains unverified against a competition-like distribution, independently of whether ω is identified (see Limitations).

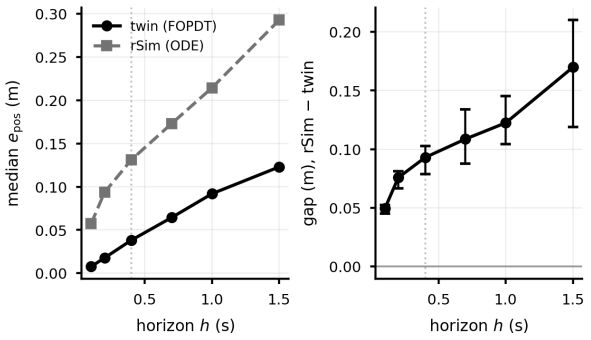


Fig. 1. Free-run position fidelity vs. horizon on translation-dominated hold-out windows, twin (FOPDT) vs. rSim (ODE engine of rSoccer), Mode A ($N = 264$ dense, 49 independent). Left: median e_{pos}. Right: gap (rSim − twin) with moving-block bootstrap 95% CI. Dotted line marks $h_{\text{ref}} = 0.4\,\text{s}$.

TABLE II
TWIN (FOPDT) VS. RSIM (ODE) ON TRANSLATION-DOMINATED HOLD-OUT WINDOWS, MODE A (WARM-UP + RIGID SE(2) RE-ANCHOR). e_{pos} MEDIANS IN METERS; GAP $= e_{\text{pos}}^{\text{rSim}} - e_{\text{pos}}^{\text{twin}}$; PAIRED NON-OVERLAPPING WILCOXON p; RANK-BISERIAL EFFECT. $N_{\text{indep}} = 49$.

h (s)	twin e_{pos}	rSim e_{pos}	gap (m)	p	effect
0.10	0.0077	0.0572	+0.0495	1.6×10^{-13}	+0.979
0.20	0.0177	0.0935	+0.0758	1.2×10^{-13}	+0.980
0.40	0.0382	0.1312	+0.0930	4.4×10^{-11}	+0.922
0.70	0.0644	0.1731	+0.1087	8.2×10^{-9}	+0.838
1.00	0.0919	0.2144	+0.1225	1.3×10^{-9}	+0.871
1.50	0.1229	0.2927	+0.1699	1.0×10^{-6}	+0.731

C. Twin vs. rSim: translational fidelity

On translation-dominated windows (Mode A), the twin's median position error is *below* rSim's at every horizon, and the gap *grows* with the horizon (Table II, Fig. 1). At the reference horizon $h_{\text{ref}} = 0.4\,\text{s}$ the gap is $+0.093\,\text{m}$ (e_{pos}: 0.038 vs. $0.131\,\text{m}$), with paired non-overlapping Wilcoxon $p = 4.4 \times 10^{-11}$ and rank-biserial effect $+0.922$ ($N_{\text{indep}} = 49$). Across horizons the effect size ranges from $+0.731$ ($1.5\,\text{s}$) to $+0.980$ ($0.2\,\text{s}$); the moving-block bootstrap CI excludes zero at every horizon and agrees in sign and order with the Wilcoxon test (no divergence flag), so the bin receives a GO verdict.

a) Robustness and second baseline: Without warm-up (Mode B, first 15 samples dropped for both plants) rSim's rest-start penalty widens the gap ($+0.282\,\text{m}$ at h_{ref}, $p = 4.4 \times 10^{-14}$, effect $+0.987$), so Mode A is the conservative primary setting; against ideal kinematics the twin is also GO on translation (gap $+0.063\,\text{m}$ at h_{ref}, $p = 8.5 \times 10^{-7}$, effect $+0.736$).

b) Heading error: Heading error carries no verdict, since the rotation and mixed bins lack scorable population (Sec. IV-B). Descriptively, the twin's median e_{ang} grows from 0.013 to $0.090\,\text{rad}$ over $h = 0.1$–$1.5\,\text{s}$, below the ideal-kinematics baseline (0.023 to $0.095\,\text{rad}$) at every horizon; against rSim, however, it is *larger* at long horizons — the position claim does not transfer to heading (Sec. VI).

D. Yaw identifiability is a function of excitation regime

A direct P1D fit of the yaw channel on high-rotation *closed-loop* competition events gives $(K_\omega, \tau_\omega, T_{d\omega}) \approx (1.35, 0.035, 0.106)$ with hold-out fit 52.6% (below the 60% acceptance threshold), and using it *worsens* free-run position because the gain > 1 amplifies accumulated heading drift; it is not promoted. A per-amplitude gain analysis shows a flat effective gain of ≈ 1.3–1.5 across the closed-loop command range, with no dead-zone knee visible in the *body*-frame yaw signal. A linear re-identification restricted to the high-rotation closed-loop regime fails whiteness (Ljung–Box rejects) and is NO-GO on the multi-horizon gate: it is *worse* than the placeholder at every horizon, pooled gap $+0.0489\,\text{m}$ at $4\,\text{s}$, $p = 3.8 \times 10^{-9}$.

a) The same structure, different data: Firmware inspection locates the dominant non-linearity as an *electrical, per-wheel* dead-zone applied after the inverse kinematics (Sec. III-E), not a body-frame Hammerstein dead-zone that a closed-loop fit could absorb. Re-deriving the commanded-yaw dead-zone from this per-wheel threshold and the robot's wheel-to-contact-radius ratio (measured from CAD, cross-checked against the firmware's implied constant) places it at 0.58–$0.60\,\text{rad/s}$ of body yaw rate – reported as an interval, never a point estimate, because the two independent geometric sources agree only to that precision. Recomputing the fraction of *commanded* yaw-rate samples in the competition logs that fall below this corrected threshold gives 94.3% pooled across all 46 splits, and 77.3% even on `shoot_to_goal`, the label richest in rotation. The closed-loop dataset simply does not excite the rotational mode enough to identify it.

We therefore identify ω from the dedicated open-loop collection (Sec. III-E), using the *same* P1D structure and identification tooling. The result is $K_\omega = 1.0016$ (95 % CI $[0.9993, 1.0040]$), $\tau_\omega = 0.0828\,\text{s}$, $T_{d\omega} = 0.0890\,\text{s}$, with hold-out fit 71.1% (65.0% / 77.2% per event), comfortably above the pre-registered 60% gate; the correlation between τ and T_d is moderate ($\rho = -0.47$), below the 0.8 alert threshold. On the multi-horizon SE(2) gate against the same placeholder, evaluated on the same closed-loop `shoot_to_goal` population used to fail the closed-loop candidate, the open-loop model is *better* at all five horizons, pooled gap $-0.1138\,\text{m}$, $p = 1.7\times10^{-16}$ – the sign of the verdict inverts relative to the closed-loop attempt on the identical gate. Table III contrasts the two rows directly.

b) Interpretation: $K_\omega = 1.0016$ lies far from the closed-loop estimate 1.3–1.35; its interval is the asymptotic, model-based one of the estimator over four training events, so it bounds estimator precision, not cross-session variability, and we read it as "near unity" in that restricted sense. The most plausible account is that closed-loop commanded yaw correlates with unmodeled disturbances (friction, contact, tactical correction) in a way that inflates the apparent gain — a property of the *data regime*, not of the DOF — though the two regimes also differ in commanded amplitude and session (Sec. VI). Re-executing both the translational and

TABLE III

YAW (ω) P1D IDENTIFICATION: CLOSED-LOOP COMPETITION LOGS VS. A DEDICATED OPEN-LOOP ROTATION-ONLY COLLECTION. SAME MODEL STRUCTURE (EQ. 1) AND IDENTIFICATION TOOLING; ONLY THE EXCITATION REGIME DIFFERS. SE(2) GATE: POOLED MEDIAN GAP VS. PLACEHOLDER (K=1, τ=0.1 s, T_d=0), NON-OVERLAPPING WILCOXON p.

	K_ω	τ_ω (s)	$T_{d\omega}$ (s)	hold-out fit
Closed-loop	1.307	0.0222	0.1179	*n/a*†
Open-loop (dedicated)	**1.0016**	**0.0828**	**0.0890**	**71.1 %**
	gate Δmedian (pool)		gate p	
Closed-loop	$+0.0489$ m (worse)		3.8×10^{-9}	
Open-loop (dedicated)	$\mathbf{-0.1138}$ m (better)		$\mathbf{1.7 \times 10^{-16}}$	

†Closed-loop row: the regime-restricted linear re-identification of Sec. IV-D (not the direct high-rotation fit quoted first there). It was scored by the earlier regime-split criterion (fit 31.9 % high-amplitude / -11.2 % low-amplitude), not a single hold-out number; it failed the physical/whiteness checks regardless.

yaw identification pipelines with identical seed, data, and toolbox version reproduces all nine parameters to 0.0 absolute difference (bit-exact), and the SE(2) gate verdict for all three DOFs is unchanged under a $\pm 21\%$ perturbation of T_d to its neighboring integer-step quantization (a robustness check, since the simulator's transport delay is in fact continuous and unquantized).

c) Negative control: joint calibration: A joint 9-parameter behavioral calibration (Bayesian optimization, multi-horizon SE(2) cost, ranking-validated analytic proxy, Spearman $\rho = 0.999$) beats the ideal baseline on the same hold-out (pooled -0.098 m, $p = 7.6 \times 10^{-10}$) but not the per-DOF model (pooled $+0.0041$ m, $p = 0.040$, wrong direction) and is not promoted – closed-loop logs do not sustain a joint behavioral fit that improves translational fidelity over per-DOF identification, the same excitation lesson that motivated the dedicated yaw collection.

V. DISCUSSION

A. Identifiability is a property of the data, not the DOF

The central finding is not that yaw is harder to model than translation, but that the *same* model structure, fit with the *same* tooling, gives opposite verdicts depending on the excitation regime of the data: on closed-loop competition logs the yaw P1D fails every axis (marginal bounds, sub-threshold fit, SE(2) gate worse than a no-dynamics placeholder), while on a small dedicated open-loop collection it passes every pre-registered gate with a gain statistically indistinguishable from unity. Framed as a property of the *data* rather than of ω: closed-loop competition commands do not excite yaw enough to identify it – 94% of commanded yaw-rate samples sit below the robot's own actuation threshold, even on the label richest in rotation – whereas the same logs excite v_x, v_y persistently (Sec. III-D); nothing in the pipeline distinguishes the DOFs, only the data does.

B. Correlation bias as the mechanism, not a coincidence

The closed-loop gain estimate ($K_\omega \approx 1.3$–1.35) is not merely imprecise; it is systematically wrong in the same direction across two independent closed-loop attempts (a direct high-rotation fit and a regime-restricted linear fit), and the open-loop estimate ($K_\omega = 1.0016$, CI excludes 1.3) refutes it as a plant property. The reading we favor is that, under closed-loop play, a human or tactical controller continuously corrects the trajectory, so the commanded yaw rate correlates with unmodeled disturbances – surface friction, contact, tactical override – in a way that inflates the apparent gain of a model fit to command-to-velocity data alone. This is a textbook closed-loop identification hazard [16]–[18], consistent with the earlier failure mode: a gain > 1 *worsens* free-run position by amplifying accumulated heading drift (Sec. IV-D).

C. Where the twin beats rSim, and why

On translation-dominated windows the twin's advantage over the ODE-based rSim engine grows with horizon (Sec. IV-C). Our reading is that the transport delay and first-order lag in Eq. 1 capture a short-horizon response that a rigid-body ODE engine without an explicit delay term structurally cannot. This should not be over-read: a residual initial-velocity mismatch also accumulates with the horizon, and rSim runs at default parameters while the twin is fitted to this robot, so the comparison bounds what an untuned reference simulator delivers on this task rather than establishing a dynamical advantage in general (Sec. VI).

D. A methodological lesson beyond this robot

We frame the yaw result as a limit of identification from competition logs, not as a repair of an earlier mistake. In the SSL, closed-loop operational data are the naturally abundant resource — league practice, as documented in its published datasets, is to record vision poses, odometry, and onboard velocities routinely [14] — whereas a dedicated open-loop campaign requires robot access outside that routine. The pattern – a model that looks unidentifiable under one data regime and identifies cleanly under another, with the discrepancy explained by a quantifiable actuation threshold rather than by the model structure – is not specific to this robot's yaw channel: "this DOF is not identifiable" claims drawn from competition logs alone should be treated as claims about the available excitation, not about the system, until checked against a regime that persistently excites the DOF in question.

VI. LIMITATIONS AND THREATS TO VALIDITY

a) Small sample; single robot and session: The open-loop yaw fit uses six linear-regime events (four training, two hold-out) from a single collection session, with no fine amplitude sweep between 0.5 and 1.0 rad/s – hence the commanded-yaw dead-zone is reported only as the interval 0.58–0.60 rad/s, with empirical bracket $(0.5, 1.0]$ rad/s. More broadly, all identification and evaluation data come from one robot and one data-collection context; we report no cross-robot or cross-session variability. Pre-registered acceptance criteria and bit-exact determinism under re-execution mitigate, but do not replace, replication.

b) Rotational fidelity of the twin remains unverified: Identifying ω resolves the *parameter*, but the twin-vs-baseline gate on competition-like data still returns no verdict for mixed and rotation-dominated windows, because the hold-out has essentially none (0 mixed, 1 rotation-dominated of 265), not because of a placeholder (Sec. IV-B). The rotational fidelity of the twin, as opposed to the yaw estimate itself, is not validated against a distribution resembling competition play.

c) The regime contrast is not a single-factor experiment: The two yaw campaigns differ not only in loop topology but also in commanded amplitude (the open-loop staircase lies entirely above the actuation threshold, whereas 94.3% of closed-loop commands fall below it) and in session, so the inverted verdict cannot be attributed to feedback alone: amplitude coverage co-varies and is a plausibly sufficient explanation, and separating the two requires a low-amplitude open-loop arm we did not collect. We also diagnose the bias of direct prediction-error identification on closed-loop data without attempting the closed-loop-aware estimators (instrumental-variable, two-stage, joint input–output) that the cited literature proposes; whether those recover the yaw dynamics from the same logs is open.

d) Kinematic constants: The firmware-implied wheel-to-contact-radius ratio ($R/r \approx 2.61$) differs from the CAD-measured one (≈ 2.699) by $\sim 3.3\%$; the likely cause is version drift between inspected source and flashed binary, unconfirmed, and we report the discrepancy rather than pick one source. An earlier body-frame yaw dead-zone constant of $0.3\,\mathrm{rad/s}$ has no recoverable provenance and is reported as inherited and superseded by the per-wheel threshold characterized here.

e) Metric, baselines, and scope: The Ljung–Box test rejects residual whiteness for both yaw fits; as pre-registered it is informative, not a gate, since the Savitzky–Golay derivative colors residuals by construction. rSim is evaluated at default parameters (a re-tuned engine could narrow the gap), and modes A/B bracket, but do not remove, the asymmetry that only the twin has an explicit delay term. All results come from a single dataset and hold-out split drawn from the same operating regime; the native grid is irregular and the free-run initial condition relies on noisy vision-derived velocity — mitigated by resampling and filtering, not eliminated.

VII. Conclusion and Future Work

A per-DOF FOPDT twin of an omnidirectional SSL robot, identified mostly from closed-loop competition logs, attains validated translational fidelity: it beats the ODE-based rSim engine on translation-dominated hold-out windows at every horizon, with a gap that grows with horizon and effect sizes of $+0.731$ to $+0.980$ (GO verdict). The yaw channel was identified from a small dedicated open-loop collection, after characterizing *why* closed-loop logs cannot support it: 94.3% of commanded yaw-rate samples fall below the per-wheel actuation threshold, and the inflated closed-loop gain estimate is contradicted by an open-loop gain near unity, which we read as correlation bias rather than plant gain. The methodological lesson travels beyond this robot: a claim that a DOF is not identifiable from operational logs is a claim about the available excitation, and should be tested against a regime that actually excites that DOF.

Future work includes validating rotational fidelity against a rotation-rich window distribution, which the competition hold-out does not provide; a low-amplitude open-loop arm to separate loop topology from amplitude coverage; cross-log and multi-robot replication; and higher-fidelity behavioral simulation as a substrate for learning-based motion planning, so far trained on simplified engines [3].

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